TFPT — Red Team / Stress Test layer

Adversarial targets A–E, the QFT round (F) and the seam round (G): try to *break* the reductions, not confirm them

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November 6, 2026 · v5.4 (rev 1284)

The TFPT document set — what is treated where

Plain language: TFPT (Topological Fixed-Point Theory) is a small discrete compiler. Two inputs — the seam constant $c_3 = \frac{1}{8\pi}$ and the carrier rank $g_{\text{car}} = 5$ — build $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ and read off the Standard-Model skeleton, the constants and the scale grammar. The series is **the introduction plus five numbered papers and four companions** (plus one standalone working note), best read in order:

0. `introduction` — reading guide, compiler closure, predictions, the dependency DAG and the single proof ledger (`verification/status_ledger.csv`). *(entry point)*
1. `tfpt_1_architecture_e8` — the two axioms, the derivation map, the EM fixed point α^{-1}, the $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ construction, the QBL theorem chain.
2. `tfpt_2_standard_model` — the SM in one φ_0-ladder formula, flavor from parabolic transport (sheet diamond, dual anchor, Q_+ cohomology), the worked closures.
3. `tfpt_3_e8_audit_bootstrap` — the seven E_8 slices as an audit raster, the cascade bridge, the Möbius bootstrap.
4. `tfpt_4_frontier` — honest status of η_B, m_p/m_e, Koide, dark matter, full QG.
5. `tfpt_5_redteam` — the adversarial audit: declared attacks, what survives, what each reduces to, and the kill tests.

Companions: R. `tfpt_research_contracts` — the open interfaces $(v_{\text{geo}}, G_{\text{net}}, F_{\text{transfer}})$ as numbered contracts; **H.** `tfpt_horizon_readouts` — Appendix H (unit-system reframe): one seam constant as the universal horizon thermal code, the Nariai/anchor surface, the resummed seam clock; **O.** `origin_theory` — why no free fundamental number remains (exact core + one honestly-typed cyclic interpretation); **S.** `tfpt_safeguards` — the verification discipline: every mechanism that defends a claim against coincidence and numerology (status calculus, no-free-pattern, the over-determination map, the firewall, frozen predictions + null model, the independent Wolfram and Lean paths, the red team).

Working notes: N1. `note_e8_gaussian_code` — the Gaussian code bridge: E_8 over $\mathbb{Z}[i]$ via Construction A over the extended Hamming code $[8, 4, 4]$, the canonical four-bit quotient $L/(1+i)L \cong \mathbb{F}_2^4$, and the G_{31} quartic companion (exact algebra, machine-certified).

You are reading document 5 (Red Team).

TFPT status at a glance — read this card the same way in every document

Two structural interfaces, one compiler. P1 is a primitive reflection-positive seam kernel with unit winding; its normalization is $c_3 = \frac{1}{8\pi}$. From this datum and the five-slot carrier $g_{\text{car}} = 5$ (axiom P2), the discrete compiler $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ is *built*, and the Standard-Model skeleton (gauge group, three families, hypercharges, the flavor matrix), the dimensionless constants ($\alpha^{-1} = 137.0359992$) and the scale grammar are read off as fixed points. The numerical readouts c_3 and g_{car} are elementary symmetric readouts of the anchor $a = (1, 1, 2)$. The algebraic core is machine-checked; physical readouts run through *named bridges* whose status is typed below. Existing reductions share the four-mark geometry and anchor $a = (1, 1, 2)$; both W-bridge premises are derived at finite level (`v1014/v1015`), modulo the external MMST identification. The additional P1 determinant-response identification is separate. The Architecture chapter "Shared origin" consolidates these scopes (`REC20261105C3`).

What is closed, and what genuinely remains. The discrete/algebraic layer is closed ([**E**]); the everything-open residual reduces to *three named interface problems*:

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Interface	Question	Status
v_{geo}	the one metrology unit ($= 1/\sqrt{G} = m/\mu$); CLOSED METROLOGY ($\mathbb{R}_+$ torsor; one dim. calibration by dimensional analysis; No-Unit v153/v725)	closed metrology [O]
G_{net}	simple-current extension $(D_5)_1 \otimes (A_3)_1 \rtimes \langle (1,1) \rangle \cong (E_8)_1$	[C]
F_{transfer}	one functor, four typed interfaces (Koide, η_B, axion, m_p/m_e)	[C]

Beside that compiler rest sits the strict-physical-TOE rest: the ten-component Rest_{TOE} / the eight gates of `TFPT.TOE.COMPLETE.01` [O] ($\text{TOE}_{\text{Complete}} :=$ AND(T1..T8); not a closure).

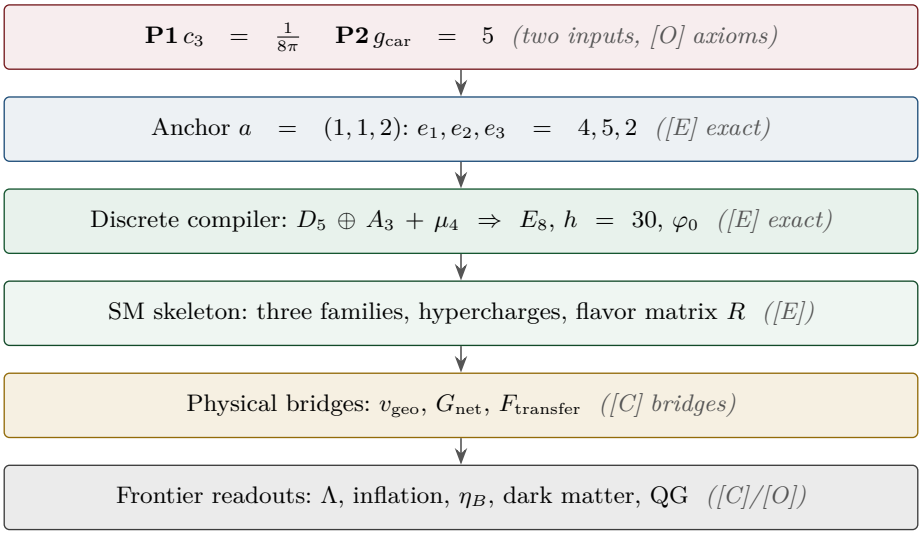
The classical covariant field equation carries no free dimensionless Newton coupling under the named QFT/entanglement premises [C]. The classical metric-sector field equation is no longer only an $R{+}R^2$ readout: the fixed-volume entanglement equilibrium (Jacobson; Faulkner et al.), run with TFPT's atoms, gives the *full* covariant Einstein equation $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$ with *both* coefficients fixed — $c_3^{-1} = 8\pi$ (no free dimensionless Newton dial; the thermodynamic origin $2\pi/\eta$ *coincides* with the geometric one $|\mathbb{Z}_2|2\pi\chi$ via $|\mu_4| = |\mathbb{Z}_2|\chi(S^2) = 4$, so c_3 is triply over-determined — anchor, geometry, thermodynamics, `v358`) and Λ from α ($\rho_\Lambda = (3/4\pi^2)e^{-2\alpha^{-1}}$, `v60`); the Einstein tensor (not Ricci) is forced by Lovelock, so matter conservation is an output (`v359`). The residual here is only the equation-of-state status (an external candidate action — Bianconi's entropic action, arXiv:2408.14391 — is quantified in `v473` without changing the typing) and the one unit v_{geo}; the global measure (C7 = `QG.AMB.01`) is now a [C] redundancy (`v369`), a certification object rather than missing dynamics, conditional on `SEAM.EQUIV.01` and Bisognano–Wichmann.

The seam-net structural theorem G_{net} (= `SEAM.EQUIV.01`: the raw seam *is* the holomorphic $(E_8)_1$ net) now carries two named routes (route split 2026-07-22, no status change in substance): the MMST route `SEAM.EQUIV.MMST.01` is *closed modulo cited theorems*; the twistor route `SEAM.EQUIV.TWISTOR.01` (Costello–Li, prepared by `CELEST.SEAM.01`) and the unconditional parent stay [O]. The lattice and index witnesses and the cited scaling and OS theorems establish the stated conditional route; they do not leave every old prerequisite untouched. The later charged MMST manuscript proves its uniform Green, edge graph, crossed-product and relaxed TEL-B estimates and retains the named microscopic local-algebra exhaustion condition for the full source. Its P1 corollary separately requires identification of the determinant-line response with the additive transitive four-channel response. The declared operational seed already supplies state and time data. The physical question concerns applicability of these theorems to the same source. The unconditional seam parent and T1–T8 keep their ledger statuses. Under the conformal-net hypotheses, rotation invariance makes the geometric deck a consequence (`v335`). The nonperturbative quantum-gravity measure `QG.AMB.01` is no longer carried as an open item: it is a [C] *redundancy* (`v369`), a certification object rather than missing dynamics, conditional on `SEAM.EQUIV.01` and Bisognano–Wichmann (and TFPT is moreover rigorously gap-decoupled from the general Euclidean-QG conformal-factor problem, margin $1.648 > 0$, `v76/v330`). So the honest residual is the one unit v_{geo} plus the typed F_{transfer} interfaces, above the cited-theorem ceiling on the seam and with `QG.AMB.01` a [C] redundancy. Everything carries a claim ID resolving to a row of `verification/status_ledger.csv` (the single source of truth); **if the text and the ledger ever disagree, the ledger wins.** The historical labels ($U_{\text{wall}}, G_{\text{metric}}, F_{\text{frontier}}$) are retained only for ledger continuity — the live residual is $v_{\text{geo}} \oplus F_{\text{transfer}}$, with the seam G_{net} closed modulo cited theorems on its MMST route (parent [O]) and `QG.AMB.01` a [C] redundancy.

*Marker key: **[E]** exact / machine-proven · **[C]** conditional (holds under named hypotheses) · **[O]** open / axiom · **[X]** falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.*

[E] exact / machine-proven [C] conditional [O] open / axiom [X] kill test

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What this note is

This is the deliberately *adversarial* layer requested in the v78 review. Its purpose is **not** to confirm TFPT but to attack the five load-bearing reductions at their weakest logical transitions. Each target is run through one fixed protocol — minimal statement, assumptions, logical chain, validity conditions, counterexample search, limiting/degenerate cases, alternative structures, provisional verdict — and produces a deliberately narrow output: where the statement works, where it could fail, and the residual risk. The layer is machine-backed by `[verification/redteam/rt_A_e8net.py]` … `[verification/redteam/rt_E_vgeo.py]`, `[verification/redteam/rt_F_qft4d.py]` and `[verification/redteam/run_redteam.py]`; a red-team check asserts an *adversarial* fact (a counterexample really exists, a hidden assumption is really needed, a firewall really holds), and the honest outcome lives in the **status** of each target, never in a green pass. The layer has since grown a seam round (Target G): the ten-test side-blind scoreboard on the alignment bit and the first *firing* kill test of the programme — the interacting straddle law (v521/v525/v528/v529), fenced at toy level — since executed as a *selector* with exactly one survivor: reflection positivity dynamically selects the bit $\delta = \pi/2$ (v534).

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Target	Minimal claim	Status	Residual risk
A	seam–Calderón measure = $(E8)_1$ net	**[C] closed mod. cited thm**	*one* statement: boundary-net holomorphy + c=8 ($\Leftrightarrow$ the index-4 inclusion); E_8 and bulk uniqueness then automatic (v83/v87/v89; Lie-level realisation v143). Now closed modulo cited theorems on its MMST route `SEAM.EQUIV.MMST.01` (the parent `SEAM.EQUIV.01` stays **[O]**): the lattice model (v367/v368) and the S_3 stack (c=8, 248 currents/1 primary, torus GSD=1, RP; v376–v379) pin it at every computable level, Lean-formalised as `FORM.SEAM.MMST.01`; only the cited MMST/OS continuum existence stays **[O]** (v336) — its 128-spinor extension leg since re-founded on 1995–2001 peer-reviewed crossed-product theorems (h_s=1$\in\mathbb{Z}$, Longo–Rehren/Böckenhauer/KLM; AGT/AMT second witness) with the realisation input reduced to invariant level (v469). The free-bulk premise is a fixed-point theorem, the cone is eliminated (cone gap = one-particle gap), and the residual factors into the A_2 net assembly + `GATE.QGEO` (v160–v165).
B	$g_{\text{car}} = 5$ forced by the Pascal rule	**survives (narrowed)**	boundary derivation of half-spinor exhaustion (degree-2 truncation)
C	$k = c_3/2 = \frac{1}{16\pi}$, $S/A = \frac{1}{4}$	**survives**	absolute $1/G$ (UV-sensitive) is the one anchor
D	ratios+products $\Rightarrow$ one scale v_{geo}	**survives (narrowed)**	CP phases ($\delta_{\text{CKM/PMNS}}$) not covered by v_{geo}
E	one scale v_{geo} carries the theory	**survives (narrowed)**	EW/reheating/leptogenesis scales + seam=Planck identification

Method

The red team treats each reduction as a hostile witness. A confirmatory script that always passes is worthless here: the layer is built so that it *may* downgrade a claim. Four outcomes are allowed — **survives** (stands as worded), **survives (narrowed)** (stands only after a silent assumption is made explicit), **reduced, not closed** (the conservative wording is the correct one), and — reached by Target A after the lattice/S_3 round — **closed modulo a cited theorem** (pinned at every computable level, only an external published existence theorem left). A fifth, **broken**, is reserved for an actual failure; none occurred on the load-bearing surface. The closest call is deliberately

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kept visible rather than filed away: the first genuinely *interacting* seam toy fires Kill-Test 2 of the OS twistor contract *at toy level* — reflection positivity breaks after the *straddle law* (Target G below) — a fenced toy result, not a target downgrade, and simultaneously the first hard selection principle the adversarial layer has produced; run as such, it keeps exactly *one* interacting member alive (v534, Target G).

Target A — the $(E8)_1$ boundary-net identification

Minimal claim. The ambient metric/projective measure is reduced to identifying the seam–Calderón boundary measure with the $(E8)_1$ lattice net, carrier subnet $(D5)_1 \times (A3)_1$. [verification/redteam/rt_A_e8net.py]

Logical chain (established). The Sugawara level-1 central charges are $c(E8)_1 = \frac{248}{31} = 8$, $c(D5)_1 = 5 = g_{\text{car}}$, $c(A3)_1 = 3 = N_{\text{fam}}$, so the conformal-embedding criterion $c_{\text{coset}} = c(E8) - c(D5) - c(A3) = 0$ holds [E]. This is *compatibility*.

Where it breaks: central charge underdetermines the net

Counterexample: $(D8)_1 = SO(16)_1$ also has $c = \frac{120}{15} = 8$, but is *not* holomorphic (four primaries $1, v, s, c$) — a distinct $c = 8$ net. Hence equal central charge is *necessary, not sufficient*: $(E8)_1$ is the unique *holomorphic* $c = 8$ net (even unimodular rank-8 lattice = $E8$), so **holomorphy (single vacuum sector, μ-index 1) is the load-bearing extra assumption**. Dropping chirality is worse: the $c = 8$ Narain family is positive-dimensional. The constructive map seam–Calderón kernel $\to (E8)_1$ and the uniqueness of bulk reconstruction are not established; the gap $\Delta_{\text{eff}} > 0$ *supports* tightness (clustering) but does not prove it.

Verdict. Keep "reduced to a rigorous boundary-net identification problem"; do *not* write "metric sector closed". The red team confirms the conservative wording. Status [C]. Residual: holomorphy proof + constructive boundary map + bulk-reconstruction uniqueness.

Target B — carrier rank / the Pascal condition

Minimal formal claim. $2^g = g^2 + g + 2$ has the unique solution $g = 5$ (Lean-verified [E]). This arithmetic is *not* attacked. The attack is upstream: why must the carrier obey the Pascal half-spinor exhaustion $2^{g-1} = \binom{g}{0} + \binom{g}{1} + \binom{g}{2}$? [verification/redteam/rt_B_pascal.py]

Where it narrows: the rule, not the arithmetic, selects 5

Perturbing the constant breaks it: $2^g = g^2 + g + c$ gives $\{5\}$ only at $c = 2$; neighbours give other or empty solution sets. The truncation *degree* is the real choice: with $2^{g-1} = \sum_{k \le K} \binom{g}{k}$ one finds $K{=}0 \to g{=}1$, $K{=}1 \to 3$, $K{=}2 \to 5$, $K{=}3 \to 7$, i.e. $g = 2K + 1$. So choosing $K = 2$ (to reach $g_{\text{car}} = 5$) is a physical postulate. It is corroborated — the $D5$ half-spinor $\dim S^+ = 2^{g-1} = 16 \Leftrightarrow g = 5$ and the family closure $N_{\text{fam}} = (2^{g-1}{-}1)/g$, $g + N_{\text{fam}} = 8$ both pick 5 — but each is itself a postulate, not a theorem.

Verdict. Theorem A's arithmetic core stands; the Pascal *selection* must be typed [O]/[C], not [E]. Residual: a boundary/seam derivation of half-spinor exhaustion.

Target C — $k = c_3/2$ and the seam-area coefficient

Minimal claim. In reduced units $k = c_3/2 = \frac{1}{16\pi}$ and Fursaev–Solodukhin gives $S/A = \frac{1}{4}$. [verification/redteam/rt_C_kc3.py]

The dimensional firewall holds

The only legal chain is

$$k_{\text{red}} = \tfrac{c_3}{2} = \tfrac{1}{16\pi} \text{ (dimensionless)}, \qquad k_{\text{phys}} = \tfrac{k_{\text{red}}}{G}, \qquad S = 4\pi\, k_{\text{phys}} A_{\text{phys}} = \tfrac{A_{\text{phys}}}{4G}.$$

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The forbidden form $k_{\text{phys}} = c_3/2$ is dimensionally false (legal only at $G = 1$). A scan of `v67/v68/v73` finds every $c_3/2$ used as the *reduced* coefficient and **zero** naked dimensionful identities; the firewall's teeth are verified against a synthetic violation. Internally consistent with $1/G$ being UV-sensitive (Sakharov/Connes, `v68`).

Verdict. Safe as worded (reduced coefficient). Status [E] (structure) + [O] for the residual. Residual: the absolute $4D$ Newton scale $1/G$ ($\propto \Lambda^2 f_2$) stays the one dimensionful anchor; nothing *dimensionless* is open.

Target D — $U_{\text{point}} \to v_{\text{geo}}$

Minimal claim. Ratios plus sector products determine the dimensionless amplitudes, leaving one common scale v_{geo}. [`verification/redteam/rt_D_upoint.py`]

Where it narrows: the bijection needs explicit hypotheses

For positive reals, (ratios, product) $\Leftrightarrow$ (individuals) is a bijection, and the TFPT sectors satisfy it (positive rational c, odd size $3 = N_{\text{fam}}$). But it *fails* off-assumption: even sector size leaves a sign ambiguity ($\{2,3\}$ vs $\{-2,-3\}$ share ratio $3/2$ and product 6); complex amplitudes leave an n-th-root branch ambiguity ($\{1,1,1\}, \{\omega,\omega,\omega\}, \{\omega^2,\omega^2,\omega^2\}$ share all ratios and product 1); a zero amplitude is degenerate. **Genuine residual:** the bijection fixes amplitude *magnitudes* only — CP phases $\delta_{\text{CKM}}, \delta_{\text{PMNS}}$ are complex and are *not* folded into v_{geo}. **Narrowed (T177 mirror,** [`verification/v561_cp_channel_identities.py`]**):** the magnitude data carry *exactly zero* phase on the TFPT sectors ($\arg \prod_i c_i = 0$ exactly for both the lepton and the up-quark triple), and the μ_3 branch group is blind precisely because its order divides $N_{\text{fam}} = 3$ (a μ_4 branch would already shift the product phase) — so the residual is a *fiber* datum (the hexagonal CM unit), not a magnitude datum. **Decided (T178 frame door,** [`verification/v564_cp_fiber_pin.py`]**):** the fiber location is now a *theorem*, not a search result — the fiber acts on the frame by a scalar on one row, so every homogeneous frame block factorises exactly (Euler), and the metric-free invariants of $(\mathbf{1}, d, n)$ have *one* generator (det; the bracket enumeration is exact): no frame-coupled scale-free $\pi/3$-carrier exists without importing a metric, and every metric import is non-canonical (a continuum of normaliser-relative values). The CP residual is *relocated with a proof*: it is the fiber datum $\operatorname{Im}\rho = \sin(\pi/3)$, and it provably cannot be made frame data.

Verdict. Holds for TFPT once four hypotheses (real, positive, fixed branch, no zeros / odd sector size) are made explicit; they are silent in `v75` and must be named. Status [E] (reduction) + [O] (v_{geo}). Residual: the CP-phase sector.

Target E — v_{geo} as the unique dimensional floor

Minimal claim. No pure-number theory can derive an absolute dimensionful scale; all scales are ratios to one unit v_{geo}. [`verification/redteam/rt_E_vgeo.py`]

Single-scale audit: exact for the certified tiers, conditional beyond

The closed compiler + protected IR readouts all reduce to (pure number)$\times v_{\text{geo}}$: $M_{\text{scal}} = c_3^{7/2}\bar{M}_{\text{Pl}}$, $f_a = (c_3/|\mu_4|)^{7/2}\bar{M}_{\text{Pl}}$, masses $= (\pi/\sqrt{2})\, c_f\, \varphi_0^k\, v_{\text{geo}}$, and $1/G$ shares the same anchor (`v68/v75`). The audit *surfaces* candidates that are not pure reductions: the seam cutoff Λ (only an *identification* seam=Planck collapses it to v_{geo}), the electroweak matching scale, the reheating temperature, and the leptogenesis scale — all frontier [C]/[O] inputs that must not be silently absorbed into v_{geo}.

Verdict. v_{geo} is the unique dimensional floor of the certified tiers (true by dimensional analysis). Full single-scale-ness is conditional on the flagged identifications staying honestly typed. Status [E] (ratios) + [O] (one scale). Residual: explicit reduction (or honest frontier typing) of the EW / reheating / leptogenesis scales + the seam=Planck cutoff.

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Target F — the perturbative 4D-QFT + scale round (v269–v275)

Minimal claim. The round published in release 5.2 — the Epstein–Glaser $S_{\rm pert}$ layer (v269/v271/v273), the PMNS Jarlskog CP strength (v270), the absolute ν-scale typing (v272), the scale over-determination (v274) and the `QG.AMB.01` roadmap (v275). [`verification/redteam/rt_F_qft4d.py`]

Two attacks land; the round survives once they are named

(1) The gravity $S_{\rm pert}$ ghost is a truncation artefact. Power-counting renormalizability is *not* unitarity: the local R^2/Weyl^2 *truncation* gives a four-derivative propagator $1/(p^2(p^2+M^2))$ with a negative-residue massive spin-2 mode (the Stelle ghost). But that is precisely the Seeley–DeWitt *truncation* of the seam KMS form factor: the entire factor $a(p^2)=e^{p^2/M^2}$ keeps its only pole at $p^2=0$ (v304), the spin-2 sector is ghost-free via the Barnes–Rivers decomposition (v370), and the nearest truncation-zero modulus runs to infinity (v380), so *perturbative* graviton unitarity is established and the ghost is exactly the truncation. The matter+gauge $S_{\rm pert}$ is unitary outright, and now closes *to all orders* as a typed Epstein–Glaser/BRST contract — read strictly as a *local perturbative construction under the stated EG/BRST hypotheses*; no confinement, mass gap, asymptotic charged states or full S-matrix follows (`QFT4D.OS.RECON.01` in `tfpt_research_contracts`) (v381, `QFT4D.EG.ALLORDER.01`): dimension-4 power-counting ⇒ a finite counterterm space, BRST nilpotency $s^2=0$ for $su(3)\times su(2)$ (Jacobi), and the seam gap ⇒ the adiabatic limit, with the imported all-order T_n existence (causal factorisation) and Slavnov–Taylor identity; the *nonperturbative* `QG.AMB.01` is then a [C] *redundancy* (v369; gap-decoupled from every TFPT readout, not a TFPT structural item, v335), a certification object rather than a new gap. **(2) The over-determination is conditional.** v274's Route-2 inverts the $\rho_\Lambda/\bar{M}_{\rm Pl}^4=(3/4\pi^2)e^{-2\alpha^{-1}}$ prediction, which is itself `LAMBDA.BRANCH.01` [C]; so the 0.11% gravity-vs-cosmology agreement is conditional on the Λ-branch (the gravity route is unconditional). **Already-honest:** $J_{\rm PMNS}$ inherits δ's [C] provenance (assembled μ_6 phase, not an M_ν output) and the ν-scale is [O] with the NO floor $\Sigma m_\nu=0.0586$ eV as the one kill test (under the carrier normalisation $y_\nu{=}y_t$ it is a one-parameter, PS-window-bracketed [O] candidate with the integer c_3-rung declined, v481 — and meanwhile *excluded* unstructured by the bracketed decision computation, v482).

Verdict. Survives (narrowed): the round stands as typed; the two narrowings above are now folded into v269 and v274. Residual: the perturbative graviton sector is now established *unitary* — the Stelle ghost is a truncation artefact (v304/v370/v380) — and the nonperturbative `QG.AMB.01` is a [C] *redundancy* (gap-decoupled from every TFPT readout, not a TFPT structural item, v335/v369); and the anchor over-determination is conditional on the Λ-branch. On the all-order footing (v381) the SM-non-unification is robust to all orders, and the *optional* carrier-Pati–Salam UV branch is *proton-safe* ([`verification/v385_uvbranch_killtest.py`]): its minimal $SU(4)$ gauge bosons mediate rare LFV ($K_L\to\mu e$), *not* $p\to e^+\pi^0$, so $M_{PS}\sim 3\times10^{13}$ GeV clears the binding bound by $\sim10^7$ — a frozen kill-test surface (κ/forbidden-rep/new-state/LFV) with *no* fake τ_p window (v265).

Under examination: an infinite-derivative narrowing ([C], does *not* close the gate). The Stelle ghost is a feature of the *local* $R{+}R^2/\mathrm{Weyl}^2$ *truncation*: the four-derivative propagator $1/(p^2(p^2{+}M^2))=(1/M^2)[1/p^2-1/(p^2{+}M^2)]$ carries the negative-residue spin-2 mode. But that truncation is the a_4 Seeley–DeWitt order of the spectral action $\mathrm{Tr}\,f(D^2/\Lambda^2)$, whose cutoff is *not* a free scheme — it is the $\beta{=}1$ seam KMS weight $f(u)=e^{-u}$ (v259, no new parameter). Keeping that exponential cutoff *un-truncated* points to a nonlocal, infinite-derivative graviton form factor $p^2a(p^2)$; for an *entire* factor $a(p^2)=e^{p^2/M^2}$ (nowhere zero) the dressed propagator $1/(p^2a)$ keeps its *only* pole at $p^2{=}0$ — no ghost (Tomboulis 1997; Biswas–Mazumdar–Siegel 2006) — whereas any *polynomial* (finite-derivative) truncation has a real zero = the ghost pole. So *if* the resummed form factor is entire, the perturbative ghost is a truncation artefact: a conditional [C] narrowing of this attack. It does *not* close `QG.AMB.01` — the trace regulator f is not yet shown to resum to an entire kinetic form factor $a(\Box)$, and a ghost-free *perturbative* graviton is still not the nonperturbative ambient measure. (`QGAMB.IDG.01`, [C]/[O]; consistent with the `SEAM.EQUIV.01` firewall.) [`verification/v304_idg_nonlocal_ghost.py`]

Sharpened by spin sector ([C]). The Barnes–Rivers spin decomposition makes this precise: of the 5+3+1+1=10 symmetric-tensor components the Stelle ghost is a *spin-2* state, so the entire form factor $a(p^2)=e^{p^2/M^2}$ cures exactly the spin-2 sector (its only pole stays at $p^2{=}0$), while the

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wrong-sign *spin-0* conformal mode ($c_{\text{conf}}(4) = -3/2$) is handled separately by the GHP contour (v334). So the full *perturbative* graviton is ghost-free sector-by-sector under the TFPT-fixed package — still [C] (it inherits the entire-analyticity assumption) and still not the nonperturbative measure. (`GRAV.SPIN2.UNITARITY.01`.) [verification/v370_grav_spin2_unitarity.py]

And the "entire" assumption is itself discharged at the truncation-tower level ([E]; [verification/v380_grav_kms_hessian.py]). The seam KMS cutoff $f(u)=e^{-u}$ (v259) gives the kinetic form factor $a(u)=e^{u}$ ($u=p^2/M^2$), which is entire and nowhere zero — so the spin-2 propagator $e^{-p^2/M^2}/p^2$ has its only pole at $p^2=0$. The Seeley–DeWitt expansion is the *Taylor truncation* of a: the $R+R^2$ order $1+u$ carries the Stelle zero $u=-1$, the Weyl^2 order $1+u+u^2/2$ the pair $-1\pm i$, and the modulus of the nearest truncation-zero grows monotonically with the order ($1, 1.41, \ldots, 3.07$ at order $1\to 8$) and runs to infinity. So the ghost is *exactly* the truncation and resummation removes it — "entire" means "do not truncate the heat kernel", upgrading the assumption above to a derived statement. (`GRAV.KMS.HESSIAN.01`; the exact off-shell curved-space Hessian identification stays the cited heat-kernel step, and it is perturbative, not the nonperturbative measure.)

Target G — the alignment bit and the interacting seam (the first shot that lands)

Minimal claim. The compiler's residual discrete carrier input is *one symmetry-lift bit* — flag transitivity of the four marks $\iff \tau = i$ (v512) — and the free-seam OS/RP structure that the Woit bridge (`WOIT.OS.TWISTOR.01`) builds on survives adversarial attack. The red-team questions are two: *can the bit be derived* (in which case calling it an input would be a hidden redundancy), and *can the OS/RP structure be broken*?

Attack 1 — derive the bit (ten failures, honestly banked). Every attack class that could compute the bit from side-blind data has been run and *killed*: the v512 equivalence web itself (no local jet sees the side bit), free RP/Θ existence (the *eighth* side-blind test, [verification/v521_seam_bit_rp_blind.py]), every Wick-computable mark-decorated state — twist/defect insertions — (the *ninth*; the free-plus-twist class is *exhausted*, [verification/v525_seam_bit_twist_blind.py]), and now the first genuinely *interacting*, non-Wick-computable dynamics (the *tenth*: the interaction sees δ massively, spectral spread 1.32, yet the mark-axis mirror maps $H(m) \to H(8-m)$ exactly — all side data mirror-equal; [verification/v529_seam_interacting_toy_fk.py]). Ten side-blind tests, ten kills: the bit is *not* derivable from any tested class and remains genuine discrete input. The flip side is a sharper ontology, not a promotion: the bit is now physically *defined* as the twist-class choice — facets #14/#15 of the (now 15-fold) web are exact equivalences both directions, the flip-axis fraction is a gauge-robust order parameter ($O = \frac{1}{2}$ at $m = 4$, 0 elsewhere), and its η holonomy is in principle interferometrically readable ([verification/v528_seam_bit_twist_class_definition.py]; [C] measurement sketch — a definition replaces no derivation, no marker moves). Since the straddle round there is also the first *positive* selection datum — not an eleventh side-blind test but a different question answered: *if* reflection positivity is demanded of the interacting seam dynamics, the surviving member is unique and it *is* the bit ($\delta = \pi/2$, positive coupling; v534 — toy-level evidence for a dynamical origin, not a derivation; the bit stays formal input).

Where it fires: the straddle law (toy level) — the first live ammunition

The minimal interacting Fidkowski–Kitaev quartic on the 16-Majorana NS seam circle (256-dim, exact) is the first adversarial probe with genuinely non-Wick-computable dynamics — and Kill-Test 2 of `WOIT.OS.TWISTOR.01` *fires at toy level*. Θ exists exactly on the interacting model ($\Theta^2 = +1$, $[\Theta, H_g] = 0$, $\Theta\Omega_g = \Omega_g$, clock-tower inversion exact), so Kill-Test 1 does *not* fire; but reflection positivity breaks in the interacting ground state for *every* $g > 0$ — inertia ladder $(37, 0, 0) \to (33, 4, 0) \to (31, 6, 0) \to (29, 8, 0)$, Gibbs state at $\beta = 2$ the same ladder, mechanism *interference* (neither crossing nor half-local terms alone break) — and the failure pattern is a law, not noise: on all 24 tested entries **RP fails exactly on quartet-straddled cuts and stays positive definite on quartet-avoiding ones** (the *straddle law*; since

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refined to the *odd-split law* — below). This is the honest reading of "first live ammunition": every previous red-team round ended in survives/narrowed; here an interacting mechanism actually breaks a load-bearing structure — at toy level, under the mandatory fence (one toy, one interaction class, [C] flat-band parent). The same law is the first *hard selection principle* for the interacting algebra: every candidate $\mathcal{A}_{\text{hol}}$ must protect RP against exactly this mechanism on quartet-straddled cuts — a concrete, checkable hurdle for the β_3/γ stages. Threat and filter are one statement. [`verification/v529_seam_interacting_toy_fk.py`]

The filter, executed (v534). The follow-up runs the straddle filter as a *selector* on the full interacting mark family (independent equivariant couplings on the four mark quartets, both signs, all admissible cuts — including the straddled clock axes 7/15 of the $\pi/2$ member that the 24-entry law had not covered). Result: reflection positivity survives for *exactly one* member × sign — $\delta = \pi/2$ with positive coupling, PD on all four cuts over the whole grid $g \in \{1/32..8\}$ with the minimum eigenvalue *lifted away* from the RP boundary; every asymmetric member dies ($s{=}+$ at $g = 1/32$, $s{=}-$ from 1/8). The law refines: asymmetric straddling kills RP, the *symmetric* straddling of the self-mirror member protects it — and the literal leading-order cone formalization is dead (the protection is nonperturbative). The filter is not merely a threat: it is a working selection principle whose unique survivor is the alignment bit — the same member that carries the twist-class order parameter $O = \frac{1}{2}$. Same fence (one toy, one interaction class, [C] flat-band parent). [`verification/v534_seam_straddle_cone.py`]

The law typed, the counterweight narrowed (v972). The refinement is now a *decidable criterion*: the Jaffe–Pedrocchi sufficient condition for Majorana reflection positivity (`arXiv:1406.1384`; Wei–Li–Xiang `arXiv:1601.01994`), made decidable on the toy, predicted RP survival on *center*-straddled cuts (2+2 mirror split; the $m = 4$ model on the clock axes 7/15) and direct measurement confirmed it ($(37, 0, 0)$ at $g = 0.5/1/2$) — a straddled cut kills RP iff its quartet split is *odd* (1+3): the straddle law refines to the *odd-split law*. The JP-certified vs measured RP confusion matrix over all 35 cells is perfect (23/12/0/0), and the μ_4-covariant quartic interaction space (dimension 464) contains *no* direction JP-certifiable on all cuts — the threat stays generic. On the counterweight side, the quartet-avoiding cut family of the clock-invariant $m = 4$ member generates the *full* even seam algebra (rank 15, dimension 32768) and carries a *positive* OS/GNS inner product at $g = 1$ (minimum $+4.4 \times 10^{-5}$, GNS 37/37, spectrum deforming continuously from $g = 0$) while the straddled control genuinely fails (-6.9×10^{-2}); the μ_4 clock maps the avoiding pair $\{3, 11\}$ onto itself exactly. Reading: the Kill-Test-2 counterweight is *narrowed* — the toy exhibits an OS-viable cut geometry selected by its own clock — *not* removed. All toy-level [E]; same fence (one toy, one interaction class, [C] flat-band parent); `WOIT.OS.TWISTOR.01` stays [O]. [`verification/v972_seam_interaction_front.py`]

Verdict. Survives (narrowed) for the bit claim: ten side-blind kills leave the bit underivable from every tested class, and the twist-class definition makes its negation measurable without moving any marker (the bit stays formal input; [C]/[O] as ledgered). For the OS/RP structure the honest outcome is fires at toy level (fenced): the free and gauge-fixed layers survive (β_1/β_2, v522/v524, including the grading, the nonsplit $\mathbb{Z}_8$ tower and the OS quotient surviving $g > 0$ constructively), while the interacting straddle threat is the named residual risk of the whole Woit route — `WOIT.OS.TWISTOR.01` stays [O], and any future $\mathcal{A}_{\text{hol}}$ that cannot pass the straddle filter is dead on arrival — while the filter's own execution (v534) shows a passing class *exists* and is unique: the $\delta = \pi/2$ member, and the odd-split round (v972) both types the mechanism (Jaffe–Pedrocchi positivity, decidable per cut) and exhibits an OS-viable avoiding-cut geometry selected by the toy's own clock — the Kill-Test-2 counterweight narrowed, not removed. Residual: the β_3/γ stages under the straddle constraint; the bit's derivation (if any) must come from outside the exhausted side-blind classes — the RP-selection route (demand reflection positivity, obtain the bit) is now the named candidate, toy-fenced.

Outcome

What the red team changed. No target broke on the load-bearing surface. The hardest gate (A) is now *closed modulo cited theorems* on its MMST route `SEAM.EQUIV.MMST.01` (the lattice/S_3 round pins it at every computable level, Lean-formalised, only an external published existence theorem left, v367/v368/v376–v379); four targets (B, D, E, F) *survive once a silent assumption is named*; one (C) survives as worded; and the seam round (G) banks ten side-blind kills on the

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alignment bit while producing the layer's first toy-level firing — the straddle law (v529), a named threat that doubles as the first hard selection principle for $\mathcal{A}_{\text{hol}}$, since executed as a selector with a unique survivor: the bit itself (v534). The net effect is not new physics but sharper typing: the package now states *where* each reduction would fail and *which* assumptions are truly necessary, exactly at the weakest logical transitions. Reproduce with `cd verification/redteam && python run_redteam.py`.

The falsifiability surface is machine-checked ([verification/v375_observatory_registry.py]). The frozen prediction registry (`freeze_file.csv`) is verified by a status-typed CI: every prediction carries a kill criterion (no orphan claims), the headline values re-derive from $\{\varphi_0\}$, and a live scorecard records each one's distance to the current data — θ_{12} vs the JUNO first run (0.3092 ± 0.0087, 0.28σ, compatible), θ_{13} vs NuFIT 6.0 NO (2.0σ, the openly-flagged most-tensioned core prediction), β_{rad} vs ACT DR6 (0.37σ, a hint with systematics pending), and $r < 0.036$ (BK18, compatible). The observatory states its own weakest point rather than hiding it.

Follow-up — the verdicts were then used as a worklist [E] ([verification/v83_e8net_holomorphic_uniqueness.py])

Two residuals were attacked directly after the red-team pass; the result is recorded in ledger rows `GATE.METRIC.04` and `CAR.PASCAL.01`.

Target A, residual 1 — closed [E]. At level 1 the number of primaries equals det(Cartan) = |fundamental group|: A_3=4, D_5=4, D_8=4, $\mathbf{E_8}$=**1**. So holomorphy (single vacuum sector, μ-index 1) is *necessary* — it excludes the same-c rival $(D_8)_1 = SO(16)_1$ ($c = 120/15 = 8$ but four primaries $1, v, s, c$) — *and sufficient*: a holomorphic $c = 8$ chiral CFT is the lattice theory of an even unimodular rank-8 lattice, of which there is exactly one, E_8, by the Minkowski–Siegel mass formula (mass $= 1/|W(E_8)| = 1/696729600$, and $|\text{Aut}(E_8)|$ saturates it). The $(D_5)_1 \times (A_3)_1$ embedding $c = 5+3 = 8$ stays *compatibility* (16 primaries vs 1). **Consequence:** the constructive map need only show the seam–Calderón boundary net is holomorphic with $c = 8$ (then E_8 is automatic), so Target A drops from *three* residuals to *two*: (i) boundary-net holomorphy $+c = 8$ ($\Delta_{\text{eff}} > 0$ supports, not proves), and (ii) bulk-reconstruction uniqueness — both still [C]/[O].

Target B — reduced [E]. The "$K = 2$ Pascal truncation" is not free: $K = (g-1)/2$ is the Pascal-row midpoint $\sum_{k \le (g-1)/2} \binom{g}{k} = 2^{g-1}$ (odd g), i.e. the half-spinor split; the carrier is the even Clifford half-spinor $\Lambda^{\text{even}}(\mathbb{C}^5) = 1 + 10 + 5 = 16 = \dim S^+$. So B's residual reduces to the single standard input "carrier = half-spinor of Spin(10)" (the Lean arithmetic core stays [E]).

Target B — a backward certificate ([verification/v219_icosahedral_mckay.py]). The choice $(|\mathbb{Z}_2|, N_{\text{fam}}, g_{\text{car}}) = (2, 3, 5)$ that B attacks is, *given* the E_8 closure, the icosahedron: by the McKay correspondence E_8 is the exceptional top of the finite-$SU(2)$ tower ($2T \to \hat{E}_6$, $2O \to \hat{E}_7$, $2I \to \hat{E}_8$), the binary icosahedral group $2I$ (order $120 = |R^+(E_8)|$) has irrep degrees $\{1, 2, 2, 3, 3, 4, 4, 5, 6\}$ equal to the affine-E_8 Kac marks ($\sum = 30 = h(E_8)$), and it contains the $2, 3, 5$ rotation-axis orders. This does *not* break B's verdict — it is a *backward* certificate (it presupposes E_8), so $g_{\text{car}} = 5$ stays the P2 interface [O]; what it removes is the impression that $(2, 3, 5)$ is an arbitrary triple, since no spherical McKay node sits above the icosahedron.

Typed honestly at this stage: A (i)+(ii) (since merged into the single keystone `SEAM.EQUIV.01`, whose MMST route `SEAM.EQUIV.MMST.01` is now *closed modulo cited theorems*, see Target A's final form below); Target D's CP-phase residual (at $\theta_{13} = 0$ the Dirac phase is undefined, so this needs a texture correction); Target E's reheating/leptogenesis scales and the seam=Planck identification. Target C is a dimensional *floor*, not a gap. **No target was promoted.**

Second follow-up round [E] ([verification/v87_bulk_uniqueness_reduction.py], [verification/v88_cp_phase_audit.py], [verification/v86_nstar_reheating.py])

Target A, residual (ii) — conditionally closed [E]: Target A = ONE residual. Bulk-reconstruction uniqueness is *not* independent of residual (i). By the algebraic-QFT classification of full 2D CFTs over a chiral net (Longo–Rehren; Kawahigashi–Longo–Müger; Bischoff–Kawahigashi–Longo–Rehren), the possible bulks are the haploid commutative Q-systems $\simeq$ physical modular invariants of Rep(A). For a *holomorphic* net Rep(A) = Vect, so the bulk pairing is *unique*. The contrast is machine-checked: the same-c rival $SO(16)_1$ (discriminant category $\mathbb{Z}_2 \times \mathbb{Z}_2$, μ-index 4) admits *six*

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modular invariants — diagonal, $s\leftrightarrow c$ swap, both E_8-extension invariants $|\chi_0+\chi_{s/c}|^2$ (the lattice glue $E_8 = D_8 \cup (D_8+s)$, h_s=1 bosonic), and two twisted pairings — so without holomorphy the $c = 8$ bulk is *multiply* ambiguous; with it, trivially unique. Hence the entire Target A reduces to the single theorem "seam–Calderón boundary net is holomorphic with $c = 8$" ([C]/[O]; ledger `GATE.METRIC.05`). The theorem also has an equivalent, more tractable *index* form ([`verification/v89_carrier_index_lemma.py`], ledger `GATE.METRIC.06`): by Kawahigashi–Longo–Müger, $\mu_A = [B{:}A]^2\mu_B$, the carrier inclusion has Jones index $[(E_8)_1 : (D_5)_1\times(A_3)_1] = \sqrt{16} = 4 = |\mu_4|$ — the glue-group order *is* the inclusion index — and the extension is the μ_4 simple-current extension (all three glue sectors are $h = 1$ currents; $248 = 45+15+64+64+60$). Since an index-4 simple-current extension of the carrier has $\mu = 16/4^2 = 1$, **holomorphy follows from the index statement**: Target A ⇔ "the seam net contains the carrier net with Jones index 4 as its μ_4 simple-current extension" — both ends constructed objects, the index controllable by Longo theory, the gap supporting finiteness. And *which* extension carries no freedom ([`verification/v92_glue_uniqueness.py`], ledger `GATE.METRIC.07`): exhaustive classification of the carrier discriminant form ($\mathbb{Z}_4\times\mathbb{Z}_4$, $q=(5x^2+3y^2)/8$) shows the extension tower is *completely rigid* — exactly two Lagrangian glues (the two chiralities, identified by the sheet $\mathbb{Z}_2$; the same two objects as the E_8-extension invariants above), and exactly one halfway extension, whose induced form $q = \{0, 0, 0, \frac{1}{2}\}$ *is* the D_8 discriminant form: the $SO(16)_1$ rival is the unique intermediate of the tower, not an arbitrary counter-model. Tower: carrier (μ=16) → $SO(16)_1$ (μ=4) → $(E_8)_1$ (μ=1, sheet pair) — nothing else exists. Target A is thereby the *bare* index statement.

Target A, final form — the Simple-Current Extension Theorem [E]/[C] ([`verification/v154_simple_current_theorem.py`]). Assembled into one statement: the carrier net $A = (D_5)_1\otimes(A_3)_1$ extended by the isotropic glue $L = \langle(1,1)\rangle$ has $[B : A]{=}|L|{=}4$, c=5+3=8, $\mu(B){=}\mu(A)/|L|^2{=}16/16{=}1$ ⇒ holomorphic ⇒ $B \cong (E_8)_1$. The *entire* adversarial residual of Target A is now the single seam-side identification "the RP seam–Calderón net *is* A" — the algebra ($B \cong (E_8)_1$) is exact and machine-checked (v89/v92/v125/v143/v148). That residual is now sharply located: the analytic core ("the kernel is quasi-free") *follows* from a free RP bulk — the boundary marginal of a Gaussian bulk is the compression of a contraction (v155), the *same* premise the area law and the EH mechanism use — and the net is *explicitly constructed and verified*: DtN = $|k|$, 16 free Majoranas, currents 248=120+128, holomorphic character $E_4/\eta^8 = j^{1/3}$ with 248 at level 1 (v156). *Final reduction (v160–v165):* the free-bulk premise is recast as a fixed-point theorem (quasi-free ⇒ all higher cumulants κ_{2n}=0, v160); the infinite Schwinger cone is *eliminated* — by Bogoliubov second quantisation the cone gap equals the one-particle gap $(2/3)^6$ (v161), whose value is forced by μ_4-equivariance (v162); and the remaining 16-dim identification factors into the two *already-open* items — A_2 net existence (an assembly of established net constructions, [C]) and R_1 = `GATE.QGEO` ("seam boundary = $\mathbb{P}^1\backslash\mu_4$") — with the irreducible core $\{\pi, v_{\text{geo}}\}$ a theorem (v165). So Target A carries *no new independent open content*: it is a unification, not an unconditional proof. The two structural residuals are then discharged to a theorem (v175): the CAR second-quantisation functor $\Gamma(t){=}\bigoplus_m \Lambda^m(t)$ makes full-cone reflection positivity reduce *for every m* to the one-particle contraction (verified on the complete 2^{16}-dim Fock space), and A_2 is an assembled, verified $(E_8)_1$ certificate; net existence and full-cone RP become [E]. Target A's single residual is therefore the seam realisation, the named keystone `SEAM.EQUIV.01` — *the raw RP seam is the holomorphic* $(E_8)_1$ *net at* $\tau{=}i$ — whose MMST route `SEAM.EQUIV.MMST.01` is now *closed modulo cited theorems* (the parent stays [O]): the explicit lattice model (v367/v368) and the S_3 stack (c=8, 248 currents/1 primary, torus GSD=1, RP; v376–v379) pin it at every computable level and it is Lean-formalised as `FORM.SEAM.MMST.01`, so the *only* residual that stays [O] is the abstract continuum existence of the scaling limit (the cited MMST/OS theorems, v336); its conformal-deck premise `QGEO.SYM.01` is its *downstream* definitional face, Lean-pinned (`FORM.SEAMEQUIV.01`, v308) to named cited steps plus the derived Recovery gap. The sprint-by-sprint reduction (v176–v302) is recorded in the changelog, not replayed here.

Target A — the keystone residual narrowed from ten further directions [E]/[C] (not promoted). A later convergent round drove the *same* single residual (the cited continuum existence, v336) down further from independent sides, while leaving `SEAM.EQUIV.01` [O]. (i) The seam edge already carries the conformal data of a chiral $c{=}\frac{1}{2}$ Majorana CFT and reproduces the c_-=8 free-fermion scaling limit on the explicit collar ([`verification/v444_seam_edge_scaling.py`]). (ii) The rigidity step is upgraded from *permit* to *force*: commuting with the order-4 μ_4 clock is *exactly* μ_4-block-diagonality (entrywise iff, uniform in N; exact commutant dimension; the order is the discriminator), kernel-hardened in Lean axiom-free as `FORM.SEAM.RIGIDITY.FORCING.01` ([`verification/v445_seam_rigidity_forcing.py`]). (iii) That clock-invariance is then *derived*, not assumed: the OS canonical transfer of μ_4-symmetric RP data inherits the clock symmetry and its

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modular Hamiltonian commutes with it ([verification/v446_seam_clock_invariance.py]), so the residual drops from an operator condition to the structural μ_4-symmetry of the seam RP data (QGEO.SYM.01). (iv) The number $c_-=8$ is re-derived from a logically independent (topological) observable — the $M=1$ collar has integer Chern $C=+1$ (with $C=0/C=-1$ controls) and the bulk–edge correspondence gives $c_-=16\cdot\frac{1}{2}\cdot 1=8$ ([verification/v447_seam_edge_chern.py]). (v) The one *external* MMST fact is pinned a further order: the edge state is quasi-free ($G^2=G$) and the four-point cross-ratio Cauchy-converges to the conformal value $\sin(5\pi/12)/\sin(\pi/12)=2+\sqrt{3}$ (against an off-scale gapped bulk control) ([verification/v448_seam_edge_fourpoint.py]). (vi) The MMST fact is pinned at the level of *uniformity*: two *independent* edge four-point cross-ratios converge to distinct conformal values ($2+\sqrt{3}$ and 2) at one N-independent $1/N$ rate ($|Q(N)-Q_\infty|\cdot N\le 4$ for both) — the empirical hallmark the abstract theorem asserts ([verification/v449_seam_mmst_uniform.py]). (vii) The number $c_-=8$ is read a *third* independent way — the Calabrese–Cardy entanglement law gives c per Majorana $=\frac{1}{2}$ ($c_{\text{Dirac}}=1.000$ vs a saturating gapped control), so $c_-=16\cdot\frac{1}{2}=8$ ([verification/v450_seam_edge_entanglement.py]). (viii) The edge CFT is named by its *operator content*: the Cardy finite-size spectrum reproduces the exact Ising primaries $h_\sigma=\frac{1}{16}$ and $h_\epsilon=\frac{1}{2}$ (L-independent), so the edge is the chiral Ising/Majorana minimal model $M(3,4)$ ([verification/v451_seam_edge_cardy_tower.py]). (ix) The $(E_8)_1$ identity is pinned on the *torus*: the character $\chi=E_4/\eta^8$ is S-invariant (one primary), T-multiplies by $e^{-2\pi i/3}=e^{-2\pi ic/24}$ ($c=8$) and is holomorphic ($c\equiv 0 \bmod 8$) ([verification/v452_seam_e8_modular.py]). (x) The last structural premise is *derived*: the four Gauss–Bonnet marks *are* μ_4 (roots of z^4-1), the form basis is μ_4-graded and the clock $z\to iz$ fixes their cross-ratio, so the residual QGEO.SYM.01 of (iii) follows from marks$=\mu_4$ plus the existing QGEO.REALIZE.01 — no new premise ([verification/v453_seam_mu4_from_marks.py]); the integer backbone of (iv)–(ix) ($c_-=8$, the Chern integers, the order-3 T-phase) is kernel-hardened in Lean axiom-free as FORM.SEAM.EDGE.CHERN.01. (xi) The certification package itself is re-founded ([verification/v469_seam_crossedproduct_route.py]): the 128-spinor extension is the *local* $\mathbb{Z}_2$ simple-current crossed product — the locality integer $h_s(SO(16)_1)=16/16=1\in\mathbb{Z}$ is exactly the Longo–Rehren criterion (LR 1995; Böckenhauer 1996; Böckenhauer–Evans 1998; KLM $\mu=4/2^2=1\Rightarrow$ holomorphic) — so the extension leg rests on 1995–2001 *peer-reviewed* subfactor theory with the AGT/AMT preprint route demoted to an independent second witness; and the realisation input is reduced from model fiat to invariants (R1′: quasi-free, gapped, class D, $c_-=8$ from P_1; computed $|C|=1$, $\nu=16$, the Kitaev sixteen-fold-way class), Lean-carried as the parallel derivation seamResidualClosed′. **None of these promotes the keystone**: the *only* thing that stays [O] is still the abstract continuum-existence theorem (v336); everything else is now corroborated, derived or kernel-checked.

Target D — the CP-phase residual quantified [E] (not closed). With the frozen leading reading $\delta=\pi/3+3\lambda_C^2=68.65°$: pull vs $\gamma_{\text{PDG}}=65.7°\pm 3.0°$ is $+0.98\sigma$ (survives). Audit, *not* promoted: the data central value sits 0.07° from the alternative coefficient reading $\pi/3+2\lambda_C^2$ ($|\mathbb{Z}_2|$ instead of N_{fam}) — a textbook look-elsewhere trap; the registry keeps coefficient 3 (REG.FREEZE.01). Decision threshold: the readings differ by $\lambda_C^2=2.88°$, i.e. 3σ-distinguishable once $\sigma_\gamma\le 0.96°$. The J-inversion is flagged magnitude-contaminated (source-scale s_{23} vs M_Z). Ledger FLAV.CP.01.

Target E — one flagged scale computed [C]. The reheating temperature is no longer a silent input: $M_{\text{scal}}=c_3^{7/2}\bar{M}_{\text{Pl}}$ [E] + standard physics give $T_{\text{reh}}=9.6\times 10^9$ GeV and $N_\star(0.05/\text{Mpc})=51.4$ [C] (ledger COSMO.NSTAR.01); honestly recorded: this slow Higgs-channel point is A_s-*disfavoured* (-11.4σ; the measured A_s requires near-instantaneous reheating), so the point is conditional on the decay channel and the frozen band $[50,60]$ stays the surface of record; the leptogenesis scale and the seam=Planck identification remain typed inputs. **Again: no target was promoted; one residual was *merged* (A), one was *quantified* (D), one input was *computed* (E).**

Target D — the CP residual geometrically *located* (not closed) [E]/[C]

The follow-up round only *quantified* the CP residual; this round *locates* it geometrically, in three consistent readings, without promoting it. **(i) A channel typing ([verification/v227_degree_exponent_channel_split.py]).** The E_8 split $248=120+128$ is a magnitude/phase split: the exponents sum to $120=|R^+(E_8)|$ (the *magnitude* channel, where the (ratios, product) $\Rightarrow v_{\text{geo}}$ bijection lives), the fundamental degrees sum to $128=\operatorname{rank}E_8\cdot\dim S^+$ (the *phase/glue* channel). The un-covered CP phases of Target D are exactly the 128 phase channel — a clean home, not a new gate; whether 128 also hosts a dark sector is Frontier, *not* asserted (this replaces the over-strong "S^- is dark matter" reading, now sharpened by v430: the other side S^- is matched $\{2,128\}$ structure, forced-disjoint from the unmapped $\{12,14,18,20,24\}$). Strengthened on the fiber by the T177 mirror ([verification/v561_cp_channel_identities.py]): the typing has an *exact* hexagonal

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realisation — $\sum_k \rho^{d_k} = 4\rho$ (arg $= \pi/3$ exactly) against $\sum_k \rho^{m_k} = 4$ (arg $= 0$ exactly) — with a parity typing forced by integer parity (all degrees even ⇒ the phase channel is sheet-*even*; all exponents odd ⇒ the magnitude channel is sheet-*odd*), so the phase-channel angle is *sheet-blind*: it carries $\pi/3$ but cannot distinguish quark from lepton (the channel typing stays [C]). **(ii) A hexagonal modulus ([verification/v220_cp_hexagonal_modulus.py]).** The seam deck is the *square* modulus (j=1728, $\mathbb{Z}/4$ clock; the kill-test selects it), which is real and carries no phase; its exceptional partner is the *hexagonal* modulus j=0 (τ=ρ=$e^{i\pi/3}$, $\mathbb{Z}/6$, CM by $\mathbb{Z}[\omega]$). The frozen leading reading $\delta = \pi/3 + 3\lambda_C^2 = 68.65°$ has leading term $\pi/3 = \arg\rho$, the hexagonal CM phase; CP lives in the $\mathbb{Z}/6$ phase *fiber* over the $\mathbb{Z}/4$ seam, not in the deck. The T177 mirror *scopes* the reading ([verification/v561_cp_channel_identities.py]): only the leading term $\arg\rho = \pi/3$ is character-covariant under the parity/sector involution; the full $\delta = \pi/3 + 3\lambda_C^2$ has *no* character (a CP-even correction added to a CP-odd angle) and is a branch/frame-relative quantity — the frozen reading itself is untouched (REG.FREEZE.01). **(iii) An orientation ([verification/v225_dual_normal_frame.py]).** The dual normal frame $d = a^\top R^{-1} = -\frac{1}{2}(1,1,-2)$ (Nariai/horizon normal) and $n = (5,-9,6)$ (first-generation selector, $n^\top R = (8,0,0)$) has oriented volume $\det(\mathbf{1}, d, n) = 21 = N_{\text{fam}}\cdot 7$, and rotating by the hexagonal unit gives $\operatorname{Im}\det(\mathbf{1}, d, \rho n) = 21\sin(\pi/3) \neq 0$ — the CP orientation *reading*, sitting exactly where the magnitude bijection fails. Narrowed sharply by the T177 mirror ([verification/v561_cp_channel_identities.py]): that reading is homogeneous of degree +1 in the frame normalisation, so the *number* $21\sin(\pi/3)$ is frame data, not an invariant; dividing by the oriented volume removes the frame identically ($\operatorname{Im}(\rho\det)/\det = \operatorname{Im}\rho$ exactly), so the invariant core is the fiber datum $\operatorname{Im}\rho = \sin(\pi/3)$ and the *sign* is a declared orientation. *Decided by the T178 frame door ([verification/v564_cp_fiber_pin.py]):* the removal is not an accident of the functional — in the metric-free (natural) calculus the frame *cannot* couple at all (one bracket generator, every scale-free ratio constant: a theorem), and every metric-declared coupling is non-canonical (the normalisers $\langle d, n\rangle = 8$, $\langle \mathbf{1}, n\rangle = 2$, $\langle n, n\rangle = 142$ give the pairwise-distinct frame factors 21/8, 21/2, 441/142, and a whole continuum q_t — normaliser-relative frame data, none of them claimed); moreover T177's two-class degeneracy is *explained* (the single-sine lemma: on crystallographic fibers $n \in \{1,2,3,4,6\}$ — exactly the two compiler clocks μ_4, μ_6 — every carrier emits the same sine, so value-based pinning is impossible by theorem) and *resolved conditional on the classical anchor*: the Jarlskog invariant carries the full character tuple $(-1,-1,-1)$, matching the sheet-odd orientation class only — the unique anchor-matching canonical carrier is the fiber-pure $\operatorname{Im}\rho = \sin(\pi/3)$ (selected, not derived). All three are [E] geometry; the physical CP *derivation* stays the Target-D [C] residual (the seam deck is not re-selected, the registry δ is untouched). **(iv) One hexagonal unit, two sheets ([verification/v231_cp_mu6_phases.py]).** The reduction is sharper still: *both* CP phases are μ_6 powers of the single hexagonal CM unit $\rho = e^{i\pi/3}$ — $\delta_{\text{CKM}}^{\text{lead}} = \arg(\rho^1) = \pi/3$ (quark) and $\delta_{\text{PMNS}} = \arg(\rho^4) = 4\pi/3$ (lepton, the neutrino phase lattice) — with $\rho^4 = -\rho$, so $\delta_{\text{PMNS}} = \delta_{\text{CKM}}^{\text{lead}} + \pi$ is the quark phase times the $\mathbb{Z}_2$ sheet ($\rho^3 = -1$); the frame orientations of (iii) are correspondingly sheet-flipped ($\pm 21\sin(\pi/3)$). So Target D's CP sector is *not* two independent phase inputs but *one* hexagonal unit read on the two sheets — the magnitude residual $3\lambda_C^2$ (quark, v88) and the physical derivation stay [C], but one of the two phase inputs is removed. **(v) The universal triality phase ([verification/v233_cp_triality_phase.py]).** In the canonical $\mu_6 = \mu_3(\text{family/triality})\times\mu_2(\text{sheet})$ factorisation, $\rho = \omega^2(-1)$ with $\omega = e^{2\pi i/3}$ the $\mathbb{Z}_3$ triality centre (the 2/3 cusp weight); since $\rho^4 = \rho^1\rho^3$ with $\rho^3 \in \mu_2$, the quark (ρ^1) and lepton (ρ^4) phases have the *same* family/triality class and differ *only* by the sheet. So the CP phase *is* the universal triality phase, sheet-split — not even a free choice of μ_6 power: the power choice of (iv) is removed, and ω is the μ_3 factor of the order-30 $(2,3,5)$ Coxeter/Milnor monodromy that builds the hull (v232). The physical CP derivation still stays [C]. **(vi) The lock as a pre-registered kill test ([verification/v320_galois_cp_relation.py]).** Because $\delta_{\text{PMNS}} = \delta_{\text{CKM}}^{\text{lead}} + \pi$ is now *forced* (one unit $\rho = \zeta_6$, the μ_4-power ρ^4, the $\mathbb{Z}_2$ sheet $\rho^3 = -1$), it is a falsifiable cross-prediction, not a free input: a measured δ_{PMNS} robustly away from 240° at DUNE/Hyper-K (beyond the sub-leading budget, $> 3\sigma$) kills the Galois-CP organisation. The relation is [E], the physical phase reading stays [C], the kill test is [X].

Global look-elsewhere closure [E] ([verification/v100_numerology_null_mc.py])

The per-target look-elsewhere traps recorded above (the $\pi/3 + 2\lambda_C^2$ alternative in Target D, the ln(46080) scale reading destroyed in [verification/v99_koide_flow_time.py]) are local instances of the global adversarial question: *could the whole scorecard be formula-fishing?* The numerology null test answers it with an explicit, declared null model run in adversarial mode: the *entire* complexity-matched formula grammar (which provably contains every scored TFPT formula — the null space includes the theory) is enumerated against conservative data windows. Joint formula-fishing probability $\prod_i p_i = 10^{-25.8}$ over

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the 13 scored observables; all 94 500 $F_{U(1)}$ variants solved, exactly one hits CODATA ($p_\alpha \le 1.1{\cdot}10^{-5}$); total $\le 10^{-30.7}$. The test is falsifiable in both directions: the negative controls (seed perturbation 1%/10% collapses TFPT's own hit count 13 → 9/6; data shuffling → 1.2) prove it has power, and the census is stable under budget/window variation (< 2 orders). Honest limit, stated as such everywhere: this is a null-model rejection *conditional on the declared grammar* — no statistical test certifies a theory; the tension observables keep their large windows and contribute nothing to the number. Details and the grammar definition: the audit note (`tfpt_3`) and the module docstring.

Structural firewall + out-of-sample seed cross-validation [E] ([`verification/v305_witness_independence.py`], [`verification/v306_seed_crossval.py`], [`verification/v465_seed_crosssector_joint.py`], [`verification/v466_seed_leptonmass_channel.py`])

The global null test answers "is the *scorecard* fishing?"; two companions answer the deeper adversarial worries "is the same integer counted many times as if independent?" and "is one shared seed dressed up as many predictions?". **Witness economy** ([`verification/v305_witness_independence.py`]): every headline integer ($|\mu_4|$=4, $g_{\rm car}$=5, $|\mathbb{Z}_2|$=2, $N_{\rm fam}$=3, dim S^+=16, rank E_8=8, 240, 248, $|2I|$=120, h=30, $|W(D_5)|$=1920, $\Omega_{\rm adm}$=48, the gap exponent 6) is a documented image of the *single* anchor a=(1, 1, 2) with π the only transcendental primitive, so the recurring $2/3{=}|\mathbb{Z}_2|/N_{\rm fam}$ and $(2/3)^6$ are *one* ratio, not many — 13 atoms from 2 free primitives (6.5× compression made explicit). The E_8 spine is genuinely overdetermined (≥ 3 structurally distinct derivations sharing only leaf inputs), while the pure-seed readouts are single-witness, so their joint surprise is scored jointly by v100, never multiplied; the negative control shows α^{-1}'s "137" is *foreign* to the anchor family (an independent witness via the $F_{U(1)}$ cubic, v3). **Seed cross-validation** ([`verification/v306_seed_crossval.py`]): treating the axiom seed φ_0 adversarially as a free dial and back-solving it from each of the five seed-grammar observables gives an error-weighted mean reproducing $\varphi_0 = 1/(6\pi) + 48c_3^4$ to 0.04%; leave-one-out predicts every held-out observable within 3σ *out of sample* (the most-tensioned hold-out is $\sin^2\theta_{13}$, $\sim 2\sigma$), and a data↔formula shuffle explodes the cross-validation χ^2 by > 100× (the test has power). **Cross-sector slice** ([`verification/v465_seed_crosssector_joint.py`]): the minimal externally-runnable form isolates the sharpest shared-origin signature — a correlation no non-shared-origin model predicts — the CMB EB/TB birefringence angle β and the quark Cabibbo angle $\lambda_C{=}|V_{us}|$ back-solve to the *same* seed within $0.01\sigma_{\rm comb}$, the three-observable one-parameter fit has $\chi^2/{\rm dof}{<}1$, and the θ_{13} channel is *declared* excluded (adding it raises χ^2 by ∼4, the documented $\sim 2\sigma$ pressure of v328), so this is the honest θ_{13}-independent statement, not a hidden dilution; a data↔formula shuffle explodes the cluster χ^2 by $>5{\times}10^4$. It upgrades no prediction (a subset of v306). **Mass-sector channel** ([`verification/v466_seed_leptonmass_channel.py`]): the same seed is read a fourth way, from a *new* experimental sector — the charged-lepton mass ratio $m_e/m_\mu{=}\frac{12}{7}\varphi_0^2$ back-solves the axiom seed to −0.11%, so the empirical over-determination now spans *five* unrelated measurement sectors (neutrino mixing, CMB, quark mixing, and now charged-lepton *mass* plus the corroborating heavy-quark m_t/m_b, −0.23%); the six-channel cluster keeps $\chi^2/{\rm dof}{<}1$. This is an *empirical* strengthening only: the formula is a source-scale ratio whose source→pole transport is the seam-gapped correction class (v393) — for m_e/m_μ it largely cancels, so the channel is added as a *consistent new sector*, not a sharper pin — and φ_0 stays fixed by the axioms (a subset of the single seed, v305 multiplicity 1). All three stay **[E]** as structural/statistical checks, conditional on the same declared grammar. **No-golden-dynamics rule** ([`verification/v305_witness_independence.py`]): the golden ratio $\phi = 2\cos(\pi/5)$ lives in the *static* carrier field $\mathbb{Q}(\sqrt{5})$ (the $\mathbb{Z}/5$ hand; minimal polynomial $x^2 - x - 1$, discriminant 5 non-square), while every *dynamic* rate is rational ($\mathbb{Q}$, the $\mathbb{Z}/6$ = sheet×family hand, e.g. the recovery factor $(2/3)^6$) — so a dynamic rate "explained" by ϕ is a cross-field false friend, and CP phases and recovery rates must run on the order-6 hand (the field firewall of v314/v315).

The *reverse* numerology audit, and what the golden ratio means [E]/[O] ([`verification/v354_e8_reverse_audit.py`])

The witness-economy box answers "every readout maps onto E_8 — is the *forward* map fishing?". The deeper adversarial question runs the *other* way: E_8 is a rich object, so how much of its structure maps onto *nothing*, and could the theory be quietly mining that richness? **The reverse audit.** Of E_8's eight Casimir invariant degrees $\{2, 8, 12, 14, 18, 20, 24, 30\}$ (exponents +1), exactly *three* feed a *primary* physical-constant readout — degree 2 (the quadratic/metric), degree 8 (the rank → $c_3 = 1/(8\pi)$), degree 30 (the Coxeter number → $g_{\rm car} = 5$ and the order-30 cycle) — and the other *five* carry no primary readout: $\{12, 14, 18, 20, 24\}$. (They do *coincide* with internal structural atoms — A_3 roots, dim G_2,

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$\det L$, $|W(A_3)|$, the anchor power sum — as v66 records; but those per-degree coincidences are mostly *unforced*, the numerology temptation that the bandwidth search v355 declines, see the next box.) So 3/8 are primary readouts, 5/8 hull overhead. **Economy, not promiscuity.** A *small fixed* set of E_8 data (rank 8, h=30, $\det K$=1, the $D_5 \oplus A_3$ sub, the 16 half-spinor) generates the *entire* readout set (gauge group, three families, hypercharges, α^{-1}, flavour, scales); the higher Casimirs are never reached into to fit. A numerological use of E_8 would be *promiscuous* (mine the rich structure for matches); TFPT is *economical* (few invariants, many readouts) and never *reaches into* the higher Casimirs to fit; where those non-readout degrees do carry content it is *forced* internal structure (the spine and the flavour det-ladder, v431), not a mined coincidence — the anti-numerology signature is precisely this economy with a forced, *non-readout* remainder. **Why the golden ratio is not numerology.** $\phi = 2\cos(\pi/5)$ sits in exactly that unmapped part: it appears only in *internal* structure (the affine-E_8 attractor eigenvalue v312, the icosian lattice v348) and in no *frozen physical-constant* readout — the sole, *conditional* exception being the axion misalignment spine angle, where $\theta_i = 3\pi/5$ is the regular-pentagon interior angle and $\cos\theta_i = -1/(2\phi)$ ([`verification/v429_axion_pentagon_phi.py`]; [C], branch-level, not a frozen prediction). Numerology means a number *fitted to an observation*; ϕ is fitted to nothing — it is the algebraic signature of the 5-fold (h=30=$2 \cdot 3 \cdot 5$), the carrier *announcing* that it is icosahedral, and even its one physical-ish appearance is forced geometry (θ_i = the g_{car}-gon angle), not a match to a measured constant. **Honest residual.** The 5/8 hull overhead is exactly that: E_8 is the minimal even-unimodular *hull* (consistency container, $\det K$=1), not the world group, so the higher Casimirs are by-products of unimodular closure — where a future higher-invariant readout could live, or genuine over-capacity (now sharpened: the five non-readout degrees are *not* diffuse slack but the forced two-family ladder $6 \cdot$ spine $\sqcup$ det-ladder, v431). Stated plainly, not hidden.

Bandwidth search of the unmapped region: the disciplined decline [E]/[C]/[O] ([`verification/v355_e8_unmapped_bandwidth.py`])

The obvious next move — "then *search* those five degrees for hits we overlooked" — is itself the trap: a structure as rich as E_8 returns a numerical match for almost anything. So the scan is run with a *strict discriminator*: a candidate counts only if it is a *forced* identity (a theorem about E_8), never a coincidence. The honest yield is mostly a *decline*, and that decline is the result. **Forced, set-level (the genuinely overlooked structure — collective, not per-number).** $\sum \deg = 128 = \dim S^+$, the spinor half of $248 = 120 + 128$ (a theorem: $\sum \deg$ = #positive roots + rank $= 120 + 8$) — the invariant "budget" of E_8 *equals the fermion content the carrier reads*; $\sum \exp = 120$ = #positive roots; $\prod \deg = |W(E_8)| = 696\,729\,600$; and the exponents $\{1, 7, 11, 13, 17, 19, 23, 29\}$ are exactly the $\varphi(30) = 8$ totatives of 30 (v223), so the unmapped degrees' exponents $\{11, 13, 17, 19, 23\}$ are part of that forced set, not free. **Suggestive but *not* a theorem (the discriminator at work).** $12 = h(E_6)$, $18 = h(E_7)$, $30 = h(E_8)$ are all degrees of E_8 (the $2T/2O/2I$ McKay ladder, v219); but "$\deg(E_8)$ contains its subalgebras' Coxeter numbers" is not a general theorem, so this appealing pattern is flagged [C], not claimed. **Declined — the numerology temptation.** The per-degree atom matches $14 = \dim G_2$, $20 = \det L$, $24 = |W(A_3)| = 4!$, $12 = |R(A_3)|$, $18 = p_4(a)$ (which v66 already flagged as fittable for 12, 18) each admit ≥ 2 readings from the frozen atoms with no forced selector, and the lone extra prime in $|W(E_8)| = 2^{14}3^5 5^2\,7$ "=" the scalaron exponent is one of many readings of 7 — all declined. **Reconciliation.** v66 (the eight degrees coincide with the load-bearing integers, expected since E_8 is the hull) and the reverse audit (v354: only $2, 8, 30$ are primary readouts) are *both* right once forced set/subchain structure is separated from unforced per-degree coincidence. The search finds *no new forced physical hit*; the disciplined decline *is* the anti-numerology outcome.

The "other side" of the seam vs the unmapped region [E]/[O] ([`verification/v430_other_side_reverse_audit.py`])

A sharp follow-up runs the audit through the *double cover*: the seam has an *other side* — the one-sided $\mathbb{Z}_2$ collar and the conjugate half-spinor S^- — so *is the leftover where the second sheet lives?* The answer is a disciplined **negative**, the sheet/deck complement of v354/v355. **The deck is the degree-2 invariant (matched).** The one-sided cover contributes $|\mathbb{Z}_2| = 2$ (the $\frac{1}{2}$ in $c_3 = 1/(|\mathbb{Z}_2|\,4\pi) = 1/(8\pi)$, v58/v216), and $2 = \min \deg(E_8)$ is the quadratic/metric — one of the *three matched* primary readouts $\{2, 8, 30\}$, never one of the unmapped $\{12, 14, 18, 20, 24\}$. **The two sheets are the 128-spinor (matched, collective).** $\dim S^+ = \dim S^- = 16$ and the spinor block is $128 = \text{rank}\, E_8 \cdot \dim S^+ = 2^{\text{rank}-1} = \sum \deg$, the spinor half of $248 = 120 + 128$; the other side S^- enters 248 as the conjugate $(\overline{16}, \overline{4})$ ($128 = 2 \cdot 64$), *not* a spare singlet. **Forced-disjoint.** The sheet/deck integer set $\{|\mathbb{Z}_2|, \dim S^+, \dim(S^+ \oplus S^-), 128\} = \{2, 16, 32, 128\}$

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has *empty* intersection with the unmapped degrees $\{12, 14, 18, 20, 24\}$; its only contact with the degree alphabet is the matched 2 (and the collective budget $128 = \sum \deg$). **Disciplined decline [O].** Each unmapped degree, "explained" from the sheet atoms, needs an *unforced* coefficient and admits ≥ 2 readings — exactly the mining the v355 discriminator declines, so there is *no* forced sheet→unmapped-degree identity. **Consequence.** This consolidates the S^--dark-matter downgrade (v227) and the no-spare-singlet WIMP no-go of the frontier E_8 scan: the other side is matched conjugate matter, the five unmapped degrees are the hull overhead, and the two do not meet. The only live "other side" question — whether the 128 phase/glue channel hosts a dark sector (v227) — concerns *matched* structure, not the unmapped five.

The unmapped five are forced structure, not diffuse overhead: the degree ladder [E]/[C] ([verification/v431_e8_degree_ladder.py])

The reverse audit (v354/v355) and its sheet/deck complement (v430) leave the five degrees $\{12, 14, 18, 20, 24\}$ with no *primary* readout. The structural follow-up asks the sharper question — *are they diffuse hull overhead, or forced internal structure?* — and the answer is the latter, an *exact* two-family decomposition $\deg(E_8) = 6 \cdot \text{spine}\{2,3,4,5\} \sqcup \big(\{2\} \cup \text{det-ladder}\{8,14,20\}\big)$. **The split is forced by 30.** The E_8 exponents are exactly the $\varphi(30) = 8$ totatives of $h(E_8) = 30 = 2{\cdot}3{\cdot}5$ (v223), hence coprime to 6, hence $\equiv \pm 1 \bmod 6$ — so the degrees (= exp +1) occupy *only* the residue classes $\{0, 2\} \bmod 6$: two arithmetic progressions, forced by the prime content of $30 = 2N_{\text{fam}}g_{\text{car}}$. **The $6k$ family is $6\times$ the spine.** Degrees $\equiv 0 \bmod 6 = \{12, 18, 24, 30\}$, divided by 6, give $\{2, 3, 4, 5\}$ — the v91 spine $\{e_3, p_0, e_1, e_2\}$ of the anchor $a = (1, 1, 2)$; $30 = 6g_{\text{car}}$ is the matched Coxeter readout, 12, 18, 24 were "unmapped." **The $6k{+}2$ family is the quadratic Casimir plus the flavour det-ladder.** Degrees $\equiv 2 \bmod 6 = \{2, 8, 14, 20\}$; here $\{8, 14, 20\} = (\det R, \det C, \det L)$ is the winding line $\det M(s, 0) = 6s + 8$ of the flavour diamond (v135), with $C = R + Q\,\text{diag}(1, 0, 0)$, and 2 is the matched metric Casimir ($8 = \det R$ is over-determined: also $\text{rank}\, E_8$). 18 **is not a holdout.** $18 = 6{\cdot}3 = 6N_{\text{fam}}$ is the spine-family member $6p_0$, not on the det-ladder ($6s + 8 = 18$ has no integer s): it changes *family*, not status, so *no* genuinely orphan degree remains. **Special to E_8.** The clean $\{0, 2\}$-mod-6 split needs both $6 \mid h$ *and* exponents=totatives(h); among $\{A_4, D_5, E_6, E_7, E_8, F_4, G_2\}$ it holds for E_8 but *fails* for E_6, E_7, D_5, so the decomposition is not a generic mod-6 triviality. **The honest v355 line.** The *arithmetic* decomposition is exact [E]; but the *functorial* claim "the $6k{+}2$ Casimir stratum *is* the flavour sheet operators" needs a representation-theoretic map and stays [C] (degrees 12, 24 still admit two canonical readings each: 6·spine vs $|R(A_3)|/|W(A_3)|$). So this is a *structure* theorem — it reclassifies the 5/8 overhead of v354 to *zero* unfamilied degrees — not a forced functorial flavour map, and it closes no gate.

The two structural integers and all of E_8 are fixed by the degree multiset [E] ([verification/v437_e8_degree_joint.py])

A final consolidation of the degree audit (of v6/v66/v355/v431), staying strictly inside the v354/v355 discipline — it adds *no* new per-degree coincidence, only one joint lock. **All of E_8 is fixed by its degrees.** From $\deg(E_8) = \{2, 8, 12, 14, 18, 20, 24, 30\}$ alone: $\text{rank} = \#\deg = 8$, $h = \max \deg = 30$, #positive roots $= \sum \exp = 120$, #roots $= 240$, $\dim E_8 = 240 + 8 = 248$, $|W(E_8)| = \prod \deg = 696\,729\,600$, $\sum \deg = 128 = \dim S^+$ — every E_8 integer the compiler uses is a function of the degrees. **The joint quadratic.** The two structural integers $(g_{\text{car}}, N_{\text{fam}})$ are the *unique* root pair of $x^2 - (\text{rank}\, E_8)\,x + (h/2) = x^2 - 8x + 15 = (x - 3)(x - 5)$: the sum of roots is the rank-fill $\text{rank}\, E_8 = 8$ (v6) and the product is $h/2 = 15 = g_{\text{car}}N_{\text{fam}}$ ($2g_{\text{car}}N_{\text{fam}} = h$, v431), and $\{3, 5\}$ is the unique positive-integer factor pair of 15 summing to 8 — so $g_{\text{car}} = 5$ and $N_{\text{fam}} = 3$ are forced *together* by two E_8 degree-invariants, not one at a time. The three primary-readout degrees $\{2, 8, 30\}$ are exactly $\{\min \deg, \text{rank}, \max \deg\}$, the invariants every simple Lie algebra distinguishes. Mirrored in Wolfram; closes no gate.

External-review residual map [E]/[C] ([verification/v182_reviewer_residual_map.py])

A careful external review (2026-06-14) raised four concerns at the [C]/[O] seams: **(A)** the mapping $2D \to 4D$ ("why does the dynamics read the masses as the Plücker norm 11?"), **(B)** the abstractness of UV gravity, **(C)** the meta-look-elsewhere of the grammar choice, and **(D)** the Koide Möbius flow (non-integer flow time $t \sim 2.84$, missing QFT generator). None adds an *independent* open gap. **A** [E]/[C]: the readout integer is exact ($11 = \sum_{k \leq 2} \binom{4}{k} = \dim S^+ - g_{\text{car}}$, the QBL boundary count, v174) and the $2D \to 4D$ step is the holographic reduction, so $A = \texttt{QGEO.SYM.01} + F_{\text{transfer}}$. **B** [E]: "gravity is the geometric readout of

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Table 1: The eight E_8 Casimir invariant degrees, audited ([verification/v66_e8_casimir_degrees.py], [verification/v354_e8_reverse_audit.py], [verification/v355_e8_unmapped_bandwidth.py]; the sheet/deck complement [verification/v430_other_side_reverse_audit.py]; the forced two-family structure [verification/v431_e8_degree_ladder.py]). Only three feed a *primary* physical-constant readout; the other five carry no primary readout but are *not* diffuse overhead — they are the forced two families $6 \cdot \text{spine}\{2,3,4,5\}$ and the det-ladder $\{8,14,20\}$, the residue classes $\{0,2\}$ mod 6 forced by $h = 30$ (v431). The genuinely forced collective content: $\sum \deg = 128 = \dim S^+$, $\prod \deg = |W(E_8)|$, exponents = totatives of 30.

deg	exp	TFPT reading	forced?	primary readout?
2	1	$\lvert\mathbb{Z}_2\rvert$	forced	yes: metric
8	7	rank E_8	forced	yes: c_3
12	11	$h(E_6) = \lvert R(A_3)\rvert$	suggestive	no; 6·2 spine
14	13	$\dim G_2$	declined	no; $\det C$ ladder
18	17	$h(E_7) = p_4(a)$	suggestive	no; 6·3 spine
20	19	$\det L$	declined	no; $\det L$ ladder
24	23	$\lvert W(A_3)\rvert = 4!$	declined	no; 6·4 spine
30	29	$h(E_8) = 2{\cdot}3{\cdot}5$	forced	yes: g_{car}
collective (forced): $\sum \deg = 128 = \dim S^+$, $\prod \deg = \lvert W(E_8)\rvert$, exp = totatives of 30				
two families (forced, v431): $\{0,2\}$ mod 6; $6k/6$ = spine $\{2,3,4,5\}$, $6k{+}2 = \{2\} \cup$ det-ladder $\{8,14,20\}$				

the boundary" *is* QGEO.SYM.01. **D** [E]/[C]: $\text{Aut}(\mathbb{P}) = \text{PSL}(2,\mathbb{C})$ is the Möbius group and the cusp flow lives on the deck $\mathbb{P} \setminus \mu_4$, so the Möbius *nature* is forced by the geometry; the non-integer $t \sim 2.84$ only locates the physical point and the missing generator *is* F_{transfer}, so $D = \text{QGEO.SYM.01} + F_{\text{transfer}}$. So A, B, D collapse onto the *two already-named* residuals; only **C** is genuinely *experiment-only*: the grammar is frozen (v84), minimal and over-determined (the anchor $a = (1,1,2)$ lands in ≥ 6 mutually-independent, not-selected-for structures), a look-elsewhere-resistant signature whose residual — correctly — falls only to out-of-sample data (JUNO, CMB-S4), never to argument. The four concerns are a unification of the residual *structure*, not a closure of the underlying premises, which stay [O]/[C].

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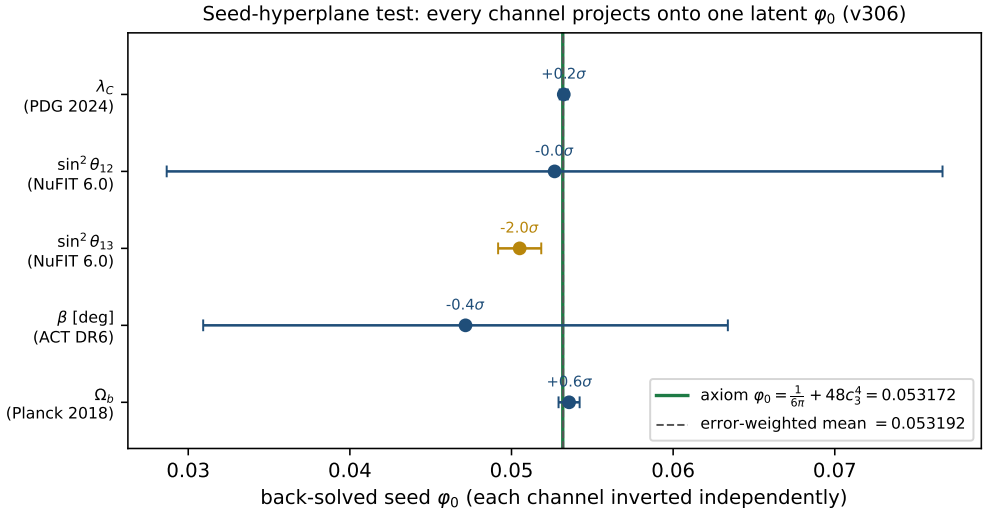


Figure 1: The seed-hyperplane test ([`verification/v306_seed_crossval.py`]): each scorecard observable is *independently* inverted to the latent seed φ_0 (with its measurement error). All five project onto the one axiom value $\varphi_0 = \frac{1}{6\pi} + 48c_3^4 = 0.053172$ (green), and their error-weighted mean reproduces it to 0.04% — one real seed, not five fits. The honest pressure point is $\sin^2\theta_{13}$ ($\sim 2\sigma$ low, a *separate* carrier-trace channel, v268; across the *mixing* channels it joins θ_{23} and $|V_{cb}|$ in the named post-hoc [O] candidate pattern v467, record unchanged); a data↔formula shuffle explodes the cross-validation χ^2 by $> 100\times$. [E].

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